

General revision

III — geometry, mensuration and data

Angles, the circle, areas, solids, and the whole of the data and probability block.

LESSON 199–202

4 × 40 MIN

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S7 WHOLE-COURSE REVIEW

LESSON CLOCK

30'

35'

35'

35'

20'

5

Angles	30 min
The circle	35 min
Area and compound figures	35 min
Solids	35 min
Data and probability	20 min
Homework	5 min

LEARNING OBJECTIVES

- Find unknown angles at a point, on a line and inside a polygon.
- Find the circumference and the area of a circle and its parts.
- Find areas of compound figures and volumes of prisms.
- Find averages, sector angles and probabilities.

TEXTBOOK

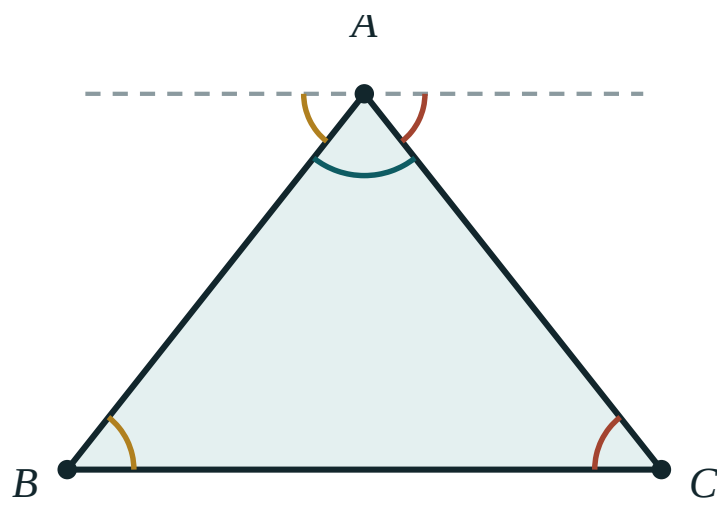
National	pp. 569–580
Cambridge	Stage 7, pp. 40–188
Workbook	Checkpoint practice 3

01

Explanation

Angles

FACT	TOTAL	EXAMPLE	UNKNOWN
angles on a straight line	180°	55° and x	125°
angles at a point	360°	90°, 130° and x	140°
angles of a triangle	180°	40°, 65° and x	75°
angles of a quadrilateral	360°	100°, 85°, 95° and x	80°
vertically opposite angles	equal	x opposite 62°	62°



the three angles fill 180°

The three angles of a triangle make a straight line

AN ISOSCELES TRIANGLE GIVES TWO FACTS AT ONCE

Two equal sides mean two equal angles. With an apex of 40° the other two share 140°, so each is 70°.

The circle

$C = 2\pi r = \pi d$ and $A = \pi r^2$

CIRCLE	CIRCUMFERENCE	AREA
$r = 5\text{ cm}$	31.4 cm	78.5 cm ²
$r = 10\text{ cm}$	62.8 cm	314 cm ²
$d = 14\text{ cm}$	about 44 cm	about 154 cm ²

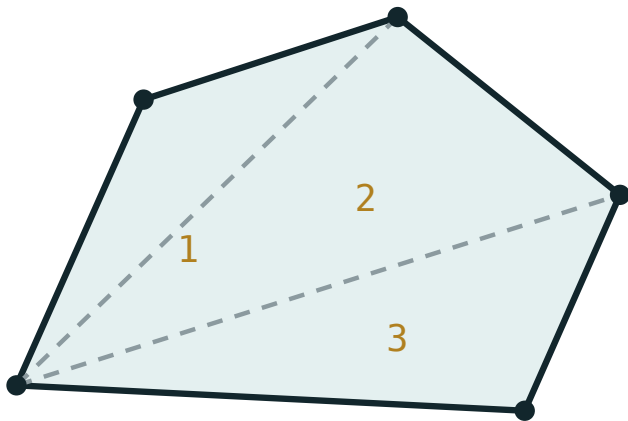
PART OF A CIRCLE	WORKING	ANSWER
perimeter of a semicircle, $r = 5$	$15.7 + 10$	25.7 cm
area of a semicircle, $r = 6$	$113.04 \div 2$	56.52 cm ²
area of a quarter circle, $r = 4$	$50.24 \div 4$	12.56 cm ²

THE PERIMETER OF A SEMICIRCLE INCLUDES THE DIAMETER

Half the circumference is only the curve. The straight edge across the bottom is part of the boundary and must be added — $15.7 + 10$, not 15.7.

Area and compound figures

FIGURE	RULE	EXAMPLE	AREA
rectangle	ab	12×7	84 cm^2
triangle	$\frac{1}{2}bh$	base 10, height 6	30 cm^2
parallelogram	bh	8×5	40 cm^2
trapezium	$\frac{a + b}{2} \cdot h$	6 and 10, height 4	32 cm^2
an L-shape	subtract the missing corner	$10 \times 6 - 4 \times 3$	48 cm^2



split into triangles, then add the areas

A compound figure cut into pieces whose areas are already known

CUT IT UP, OR FILL IT IN

Every compound figure is either a sum of simple pieces or a large shape with pieces removed. Draw the cutting lines on the diagram before writing any numbers.

Solids

SOLID	VOLUME	SURFACE AREA
$5 \times 4 \times 3\text{ cm cuboid}$	60 cm^3	94 cm^2
cube of edge 4 cm	64 cm^3	96 cm^2
prism, cross-section 12 cm^2 , length 10 cm	120 cm^3	—
cylinder $r = 5, h = 10$	785 cm^3	471 cm^2

SOLID	FACES	EDGES	VERTICES
cube	6	12	8
triangular prism	5	9	6
square pyramid	5	8	5

ONE RULE COVERS EVERY PRISM

Volume is the cross-section area times the length, whether the end is a rectangle, a triangle or a circle.
Surface area is the total of the pieces of the net.

Data and probability

TASK	EXAMPLE	ANSWER
mean	2, 5, 5, 6, 12	6
median	the same set	5
mode and range	the same set	5 and 10
sector angle	12 out of 36	120°
reading a sector	90° of 200 people	50 people
probability	$P(\text{even})$ on one throw of a die	$\frac{1}{2}$
the complement	$P(\text{not red})$ with 4 red in 10	$\frac{3}{5}$

A SECTOR ANGLE NEEDS THE TOTAL; A PROBABILITY NEEDS THE WHOLE SET

Both are shares. The commonest lost mark in this block is using the frequency where the total was needed, or the other way round.

02

Worked examples

EXAMPLE 1 A circle has radius 6 cm. Find the area of a semicircle on it, taking $\pi = 3.14$.

① $A = 3.14 \cdot 36 = 113.04 \text{ cm}^2$.

② Half of that.

③ 56.52 cm^2 .

Taking $\pi = 3.14$ ✓

ANSWER

56.52 cm^2

EXAMPLE 2 An L-shape is a 10 by 6 cm rectangle with a 4 by 3 cm corner removed. Find its area.

① $10 \cdot 6 = 60 \text{ cm}^2$.

② $4 \cdot 3 = 12 \text{ cm}^2$ removed.

③ $60 - 12 = 48 \text{ cm}^2$.

Draw the cut before calculating ✓

ANSWER

48 cm^2

EXAMPLE 3 A pie chart of 200 people has a 90° sector. How many people, and what percentage?

① 90° is a quarter of the circle.

② $200 \div 4 = 50$ people.

③ $90 \div 3.6 = 25\%$.
A quarter either way ✓

ANSWER

50 people, or 25%

03 Interactive model

Set one question from each of the five sections as a timed round; the switching between topics is the part of an annual paper that catches people out.

Geometry, mensuration and data in eight questions

Angles at a point total:

90°

180°

270°

360°

40° , 65° and x in a triangle gives:

65°

75°

85°

105°

The circumference of a circle of radius 5 cm is:

15.7

31.4

78.5

10

Its area is:

15.7

31.4

78.5

25

The perimeter of a semicircle of radius 5 cm is:

15.7

25.7

31.4

20

A trapezium with parallel sides 6 and 10 and height 4 has area:

24

32

40

64

A cylinder $r = 5$, $h = 10$ holds:

157

314

471

785

$P(\text{not red})$ with 4 red in 10 is:

$$\frac{2}{5}$$

$$\frac{3}{5}$$

$$\frac{1}{2}$$

$$\frac{6}{4}$$

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Angles on a line	Yoyiq burchak	Развёрнутый угол
Angles at a point	To'la burchak	Полный угол
Circumference	Aylana uzunligi	Длина окружности
Area of a circle	Doira yuzasi	Площадь круга
Compound figure	Murakkab shakl	Составная фигура
Prism	Prizma	Призма
Sector angle	Sektor burchagi	Угол сектора
Probability	Ehtimollik	Вероятность

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

Vertically opposite angles are:

supplementary

equal

complementary

unrelated

The area of a circle is:

$2\pi r$

πr^2

πd

$2\pi r^2$

A compound figure is handled by:

guessing

cutting it into known pieces

measuring

ignoring part of it

The volume of any prism is:

base area $\times$ length

all edges multiplied

surface $\times$ 3

base $\div$ height

A sector angle is found from:

the frequency alone

the frequency and the total

the total alone

the mean

A probability of 1.4 means:

a likely event

a mistake

a certain event

a percentage

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Angles on a straight line total

ANSWER 180°

2. Angles at a point total

ANSWER 360°

3. The angles of a triangle total

ANSWER 180°

4. The angles of a quadrilateral total

ANSWER 360°

5. The circumference of a circle of radius 5 cm

ANSWER 31.4 cm

6. The area of a rectangle 12 by 7 cm

ANSWER 84 cm^2

7. The mode of 2, 5, 5, 6, 12

ANSWER 5

Medium · the standard question

7 PROBLEMS

1. 55° and x on a straight line

ANSWER 125°

2. 90° , 130° and x at a point

ANSWER 140°

3. 40° , 65° and x in a triangle

ANSWER 75°

4. The area of a circle of radius 5 cm

ANSWER 78.5 cm^2

5. The area of a triangle, base 10, height 6

ANSWER 30 cm^2

6. The area of a trapezium, sides 6 and 10, height 4

ANSWER 32 cm^2

7. 12 out of 36 as a sector angle

ANSWER 120°

Hard · stretch and reason

7 PROBLEMS

1. The base angles of an isosceles triangle with apex 40°

ANSWER 70° each

2. The circumference of a circle of diameter 14 cm

ANSWER About 44 cm

3. The perimeter of a semicircle of radius 5 cm

ANSWER 25.7 cm

4. The area of a semicircle of radius 6 cm

ANSWER 56.52 cm²

5. The volume and surface of a cylinder $r = 5$, $h = 10$

ANSWER 785 cm³ and 471 cm²

6. 200 people, a sector of 90°

ANSWER 50 people

7. $P(\text{not red})$ with 4 red in 10

ANSWER $\frac{3}{5}$

07 Homework

Homework — 5 tasks

Write the geometric fact or the formula beside every line of working.

- Find x when x , 70° and 45° are the angles of a triangle, and again when they are angles at a point with 115° .
- Find the circumference and the area of a circle of radius 7 cm, taking $\pi = 3.14$.
- Find the area of an L-shape made from a 12 by 8 cm rectangle with a 5 by 4 cm corner removed.
- Find the volume and the surface area of a cuboid 9 cm by 4 cm by 3 cm.

5. A pie chart of 45 pupils has a group of 15. Find the sector angle, and find $P(\text{odd})$ for one throw of a die.